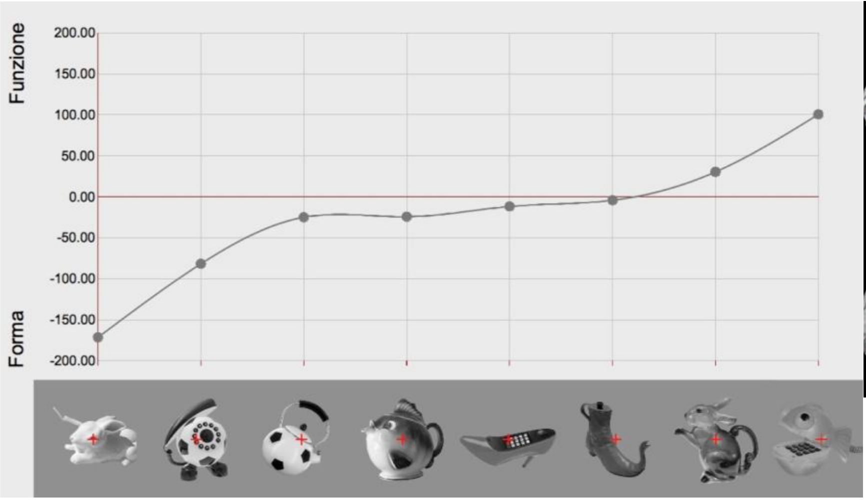
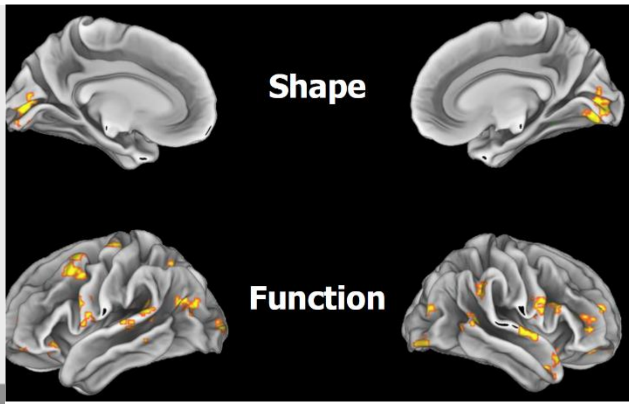
stimulus, if you look at which features are represented according to the behavioural description, you see that information about shape is just limited to the ventral stream, but when you talk about something more related about function and action, there is a broader engagement of the whole brain.





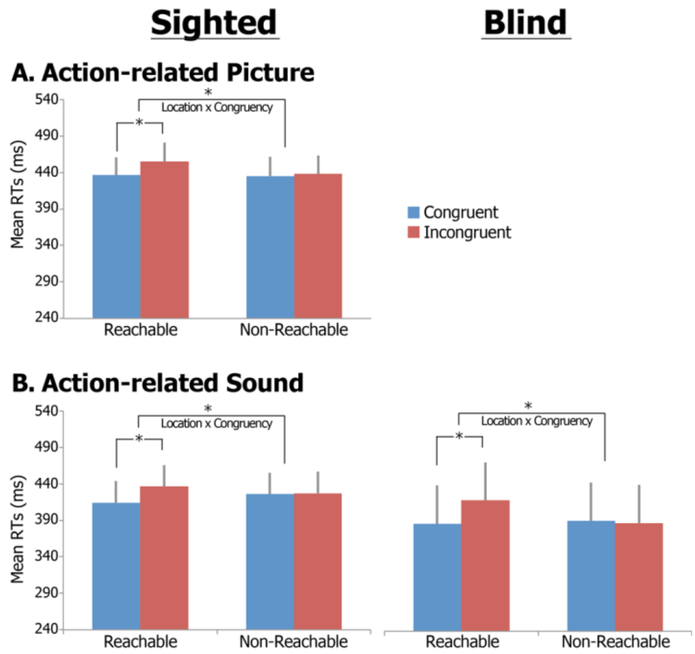
The third pathway is much more related to the superior temporal sulcus involved in social perception (recognizing facial expression, aspects useful for social interaction), a theory of just 1-2 papers. Obviously, all the information of the two pathways is highly integrated constantly, to get more complex and high-level representation.

Supramodality

Neuroscience started to consider other sensory modalities. Do we really need vision to capture all different pieces of information? When we listen to sounds that we know, our ventral stream is processing even auditorily the sound, but without being aware our brain is also processing the action that produced the sound, we do not truly need vision to make sense of the surrounding world. It seems like that several features are not actually depending on visions, sight has always been regarded as the most important sensory modality, and more than 50% of the brain is devoted to process vision, nonetheless all those who are visual deprived since birth appear to show most of their perceptual and social skills similar to those found in other individuals. There are no superior skills, they simply rely on other sensory modalities and some kind of external world representation are almost similar with some difference.

It's important to mention the spatial alignment task, which is something that can be done when you create a sort of discrepance in congruency between the vision and motor command. You can have a conditional incongruency, so the command is to perform the action with the right hand, but the ball is on the left, this creates a small delay in the response. The ball can also be in a reachable distance or not reachable, if the ball is in a space that is not reachable this effect disappears. When the object is outside our space of action, we are not processing it anymore.

The first attempt was to translate the task in an auditory one, so people were blindfolded and the ball was bouncing on the left or the right. Some individuals could present the same effect on the auditory condition, but also congenitally blind people. So blind people that never experienced vision have a sort of representation of the space around them.



You can also do the same experiment with or without an agent, that is present on the side of the table, and he just helps you reaching the ball, he's making that action possible, and the ball is in his space of action, and my space of action is the same. Important for next week lesson.

In blind people (because of defects of the eyes) we have a reorganization of the visual cortex, being it a little bit thicker and less myelinated with respect to typical individuals. But the major alteration we find are related to the visual pathways; all of them, from the thalamus back to the occipital cortex appear to be reduced and atrophic, since there is no input from the retina. There is obviously a reduction in the visual fibres of the corpo callosum, and also there is a reduction in the lateral geniculate nuclei.

But what happens to V1? Sighted people rely on primary visual cortex; while providing auditory input they respond with auditory primary regions. In blind people the primary visual cortex tends to respond to non-visual modalities, something that historically has been called cross-modal plasticity (one modality is invading the other brain areas). We also found out that the recruitment of the occipital cortex was correlated with the performance, appearing to be independent thus responding both to tactile and auditory, also across different tasks. This obviously made people wonder on the functional meaning of this. The functional significance of these changes remain unclear.

A first hypothesis is that this brain area in blind people is actually starting to do something important in comparing, a hypothesis that is truly compensatory. There is piece of evidence demonstrating that if you inhibit or you have a damage in the occipital cortex, blind people have the ability to read blind, so it looks like a sort of compensatory task could be there. Nonetheless, even this kind of compensatory role appears to be very limited, and the are for language and word representation is indeed very close to primary visual area in occipital cortex.

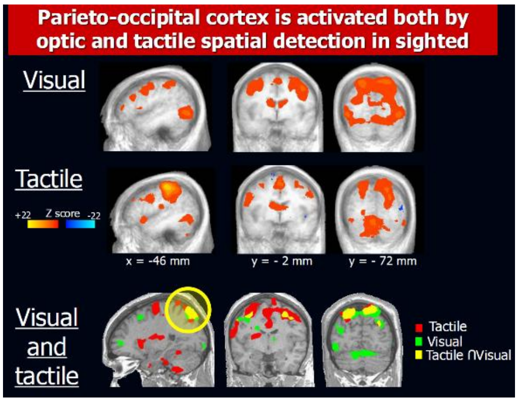
Other people think that V1 becomes pluripotent, starting to do high level functional and semantic processing, we do not believe it. In nature what is pluripotent is not highly specialized, a cell is pluripotent and is not functionally specialized in tissue, it can develop. It's very difficult that pluripotent is becoming highly productive.

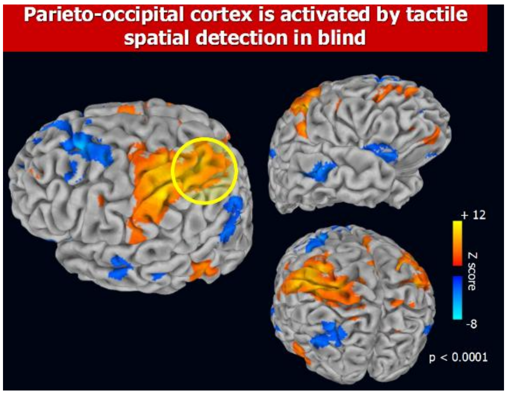
Others think that the recruitment of V1 is a little bit aspecific, that it just reflects top-down (attention) modulation. But V1 is still receiving piece of information from the other sensory modalities, since there is plenty of evidence that V1 is actually perceiving information from visual cortex. All the ventro-temporal regions are conveying information to V1, even if it is not sending anything back, there may be connections with other sensory modalities to V1.

These two perspectives are very interesting, because you are telling us that in a multisensory perspective one region when gets specific sensory input is also sharing this information with other modalities, and in a multisensory world like ours it makes sense that when we get an auditory input the visual system should be synchronized and rephase in order to respond to respond to the multisensory evaluation.

It is likely that V1 is responding somehow in an aspecific way, it is not a compensatory plastic response, but there is no clear demonstration. Also, one other thing that we have to keep in mind that different brains may have different weights of the different modalities (the bat brain is highly representing the auditory area). In rats that have very poor vision, even though it is useless they still keep a small V1 region.

In a study, subjects had to recognize tactily shoes, bottles and faces. With visual recognition, you can see a nice activation of the occipital region and of the inferior temporal cortex. Blindfolded you recognize all the elements, you have the activation of the sensory motor cortex and also of the inferior temporal cortex; vision is green, tactile is red and yellow is overlapping. So, even if your eyes are closed, when you touch an object, you recall the visual image. Blind people are different, because they are used to recognize objects touching them, so their recruitment of the sensory motor cortex is much smaller. But this area is present here, so clearly it looks like that the ventral extrastriate cortex shows category specific response on the tactile detection of objects, while the cortical area is involved in visual discrimination of objects, but also in tactile discrimination of it.

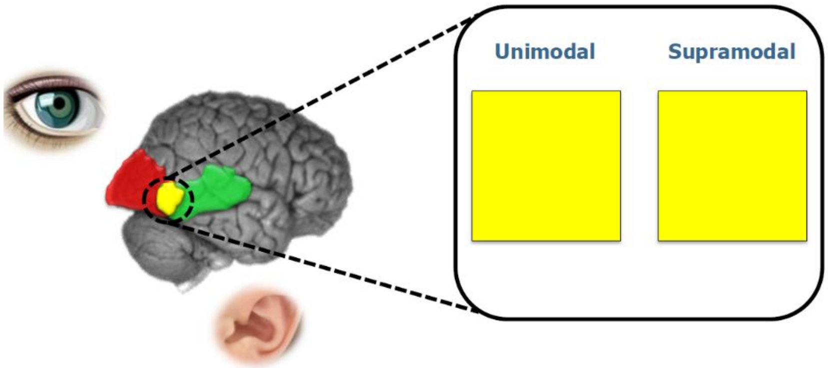




Moreover, doing the analysis showed before of category related patterns even in blind individuals we are able to get bottle, shoes and face response, but they can't really recognize face that they don't know from the touch. One of the issues when you try to recognize a face tactily is that you are not able to reconstruct the configuration of it, but only the separate elements, and this is called gestalt (the ability to bind together elements into unique percept). While you have specific category patterns for objects, it is a little bit ambiguous stimulus, nonetheless to our purpose the fact that blind people, that never got visual experience, have this type of response, is telling us that this response is not due to visual imagery but more important that visual based experience is not necessary for the development of the functional architecture of the ventral stream.

After few years, there has been several experiments about all different brain areas. And f-MRI studies in blind people told us that certain regions appear to be specifically related to capturing, for example, motion discrimination either tactily or visually. Visual and tactile modalities project on the dorsal stream, so not just the ventral but also the dorsal, and we can see in blind the same spot.

Blind people are an example of these overlapping elements, and this is actually something that we have been colling supramodal. For some individuals that for instance perceive motion for objects either visually or tactily, there is an overlapping region activated. In blind people you have the same type of recruitment, so if cross modal is something that happens because of the lack of vision, supramodal is something that takes place despite the lack of vision, something that appears to be there even without sensory input and experience.



Most of the brain appears to be supramodal for several aspects, not just motion discrimination in green or object sight in red, but also for some other like spatial orientation, mathematics. So, all the ventral and the dorsal pathway and several regions on the pre-frontal cortex, appear to be independent of the sensory modality, overlapping. Task specific but modality independent.

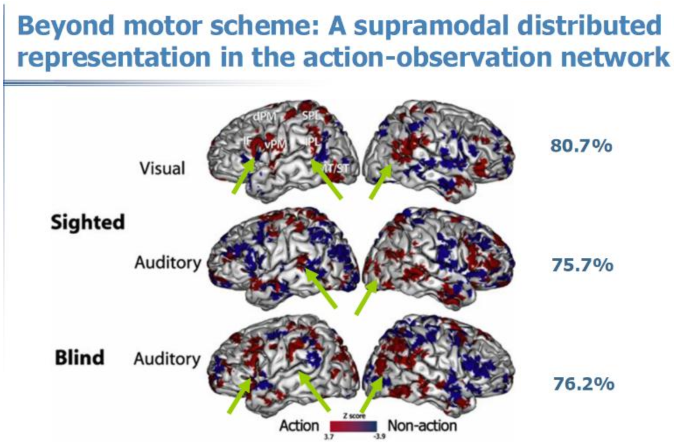
It looks like the retinotopic organization in blind is preserved as spontaneous activity, retinotopic organization is actually maintained in blind people.

So, it seems like that because of the lack of the sensory modality individuals undergo a plastic organization, something that is called crossmodal plasticity, even if we still don't know how this occurs, and brain organization is primarily driven by task specific but modality invariant computation, something that can be referred to as supramodal. The idea is that the kind of information is shared across typical developed and sensory deprived individuals, that is that when we say that blind people are differentially able to process information is true from a neuroscientific perspective, so they actually are able to differentially perceive the world, but represent the external world in the same way as a sighted individual does.

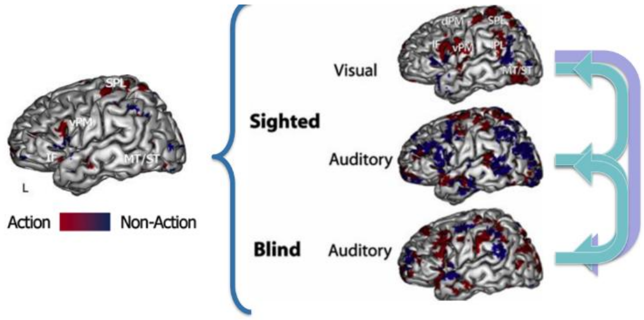
The main idea is that truly these aspects are overlapping. One main criticism that could exist is that being an overlapping region of different voxels, the voxels can be formed of different neurons of two unimodal, either tactile or visual, or there are neurons that are processing information conjointly? If we are looking at discrimination of action vs. inaction and we have 3 experimental conditions:

- Sighted individuals doing the task visually.
- Sighted individuals doing the task auditorily.
- Blind people doing the task auditorily.

The ability to discriminate is well above the 50% chance and what you can do is really get a classifier into a unique independently of the 3 modalities and what you are doing is to define regions, that are typically useful for discriminating action, with which you can train your machine learning tool on the visual stimuli and then test on the auditory stimuli and then get significant ability to discriminate action vs. non-action. And this is a sort of demonstration that this data is truly representative and there is plenty piece of evidence that are showing this aspect.



A 'machine learning' classifier discriminates a supramodal and distributed functional representation of action



- The representation of the same stimulus category (action) is accurately recognized within and across different sensory modalities (visual, auditory or motor) and experimental groups (sighted and congenitally blind): a truly supramodal functional representation of actions

Even if you go on other topics, like conceptual representation, and you have like 8 categories, and you ask blind people to produce semantic description of those, you find a very high correlation between the description of sighted and blind individuals and yet characterized by domain specificity. If you give them a banana the first thing they say is that is a fruit, so the domain is the first aspect. Blind people rely a lot on synesthetic aspects.

What happens if you restore vision? There may be function that we are able to restore, and others that still need time to develop. People whose vision is restored (kids in Africa), for many weeks are not able to recognize face, they need experience, they rely in voice to recognize a person. So, there are features (low-level features of perception) that are already present in the brain, like shape and biological motion, and this is something that can be easily restored, complex stimuli indeed, like faces, require experience. Our brain is a mixture of experience and innate features, and there is no one dominant aspect. One important thing is that clearly the brain is able to process and to develop its functional architecture independently of a sensory experience, like blind people.

Motor and action

We should depict the brain as a system that has a specific aim, collecting piece of information from the periphery in order to enact some sort of behaviour, and obviously motor behaviour action is a sort of main outcoming of brain processing. We have been actually extending our idea of how our brain get information from the external world and try to make sense and interpret this. Our brain specifically selects information that has to be collected, and the way it interprets it is always something that relates to somehow how they can modulate consequent behaviour.

One thing that is still a question, is how the brain organizes and represents simple motor element into a finalized behaviour. The way the brain represents action is something that is struggling several disciplines. If you think on how to provide commands, we have several redundant piece of information that represents most of the time noise for our purpose, and even we try to classifiy